

FIN — Release 10.27

Research Programs P295–P308

Legacy physics revisited:
positive path mixtures, strict spectral completion,
and physical-role transfer

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Central result. The legacy hyperbolic envelope is an exact positive mixture of exponential distance attenuations. The strict envelope, phase signs, and spectral order lie outside positive fixed-phase and monotone scalar-completion classes.

New boundary objects. The Typed Legacy–Strict Completion Span and the Role-Transfer Certificate separate mathematical completion from physical interpretation. No historical electroweak, electromagnetic, or gravity role currently passes the certificate.

Repository. <https://github.com/hyconiek/Fractal-Nadsoliton-Theory>

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Publication statement

This monograph reports all executed Programs P295–P308 and recommends Programs P309–P322. It corrects historical legacy-kernel calculations, preserves the legacy/strict split, and keeps mathematical proof, synthetic evidence, scoped refutation, and missing external evidence distinct. QW-2191, full bridge completion, physical-role transfer, dimensional emergence, laboratory validation, L_{total} , and Theory-of-Everything closure remain open or blocked.

Contents

Publication statement	i
Confidence convention	1
1 Executive summary	2
2 Scope, ontology, and kernel split	5
3 Reconstruction of the historical legacy-to-physics chain	6
3.1 What the old construction actually establishes	6
3.2 Corrections to the historical diagrams	7
3.2.1 Cosine zeros	7
3.2.2 Path-counting exponent	7
3.2.3 Inconsistent parameter freezes	7
3.2.4 Simulation is not measurement	8
3.3 Historical physical-role formulas	8
3.3.1 Electroweak-angle benchmark	8
3.3.2 Electromagnetic benchmark	8
3.3.3 Gravity-hierarchy benchmark	9
3.3.4 Dimensional scales	9
4 Exact legacy–strict decomposition and its limit	10
5 New theoretical objects O54–O61	12
5.1 O54 — Spectral Observation Signature	12
5.2 O55 — Bernstein Spectral Completion	12
5.3 O56 — Operational Prediction Pseudometric	12
5.4 O57 — Positive Path-Scale Measure	13
5.5 O58 — Scale-Orbit Prediction Quotient	13
5.6 O59 — Role-Transfer Certificate	13
5.7 O60 — Pointed Operational Kernel Package	13

5.8	O61 — Typed Legacy–Strict Completion Span	14
6	Program results P295–P308	15
6.1	P295 — Multi-probe operator-valued Stieltjes recovery	15
6.2	P296 — General staged elimination	15
6.3	P297 — Complex-unitary compilation of the strict current POVM	16
6.4	P298 — Scalar complement and Bernstein completion no-go	16
6.5	P299 — Multitime memory ledger	17
6.6	P300 — Positive path-mixture and fixed-phase no-go	17
6.7	P301 — External P241 intake	17
6.8	P302 — Sequential detector design with nuisance tomography	18
6.9	P303 — Local strict-fingerprint rarity	18
6.10	P304 — Adversarial finite-time equivalence	19
6.11	P305 — Intervention-assisted bridge-coordinate discovery	19
6.12	P306 — Physical reservoir evidence gate	20
6.13	P307 — Scale orbit and legacy-role invariance	20
6.14	P308 — Pointed role-transfer theorem	21
7	Cross-program synthesis	22
7.1	What survives from the old legacy intuition	22
7.2	What does not survive	22
7.3	The new bridge hypothesis	22
7.4	Why the old constants must bifurcate	23
7.5	Deepest present interpretation	23
8	Failed approaches and impossibility boundaries	24
9	Recommended Programs P309–P322	25
9.1	Recommended immediate sequence	25
10	Reproducibility	27
10.1	Executable artifacts	27
10.2	Verification command	27
10.3	Scientific boundary	27
11	Final answer	28
12	References	29
	Reproduction certificate	30

Confidence convention

- **[Proven]** — an analytic identity, exact finite theorem, or machine-checked statement in the explicitly declared class.
- **[Strong evidence]** — a reproducible numerical or synthetic audit with frozen inputs, controls, metrics, and stop rules.
- **[Moderate evidence]** — a robust result that still depends on an additional equivalence, source, or modeling assumption.
- **[Speculative]** — a proposed object or research direction without a sufficient proof.
- **[Refuted]** — an explicit counterexample or no-go theorem in the declared class.
- **[Blocked by external evidence]** — the executable gate was reached, but independent laboratory data, custody, calibration, or preregistration were absent.

A mathematical proof is not evidence that nature realizes its hypotheses.

Chapter 1

Executive summary

Programs P295–P308 were executed as a deliberately asymmetric comparison: the historical legacy kernel was reconstructed together with its old connections to physical constants, while the strict kernel was treated as the current finite spectral operator. The kernels were not silently identified, and no historical physical role was transferred to strict.

The central conclusion is:

The old legacy research contains a mathematically meaningful attenuation mechanism, but its maps to electroweak, electromagnetic, gravitational, and Planck-scale physics do not factor through a proved operator observable. The strict kernel cannot be obtained from that attenuation mechanism by positive damping, unchanged phase, or monotone scalar functional calculus. A viable bridge must therefore be typed, non-scalar, operationally testable, and separately equipped with a physical-role certificate.

The reconstructed legacy attenuation

$$h_L(d) = \frac{1}{1 + \beta d}$$

has the exact representation

$$h_L(d) = \int_0^\infty e^{-u} e^{-\beta d u} du.$$

It is therefore a positive Laplace mixture of exponential distance attenuations. **[Proven]**

The strict attenuation

$$h_S(d) = \frac{1}{1 + d^{1.8}}$$

is not completely monotone: its second derivative is negative for sufficiently small positive d . Hence it cannot be generated by the same positive distance-mixture mechanism. The unchanged legacy phase also has the wrong sign at four of the six nonzero Z_{12} distances. The best positive fixed-phase fit has relative residual 0.48169. **[Proven scoped no-go]**

The obstruction persists at operator level. Scalar complement completions force spectral flat bands with multiplicity incompatible with the strict generator. Moreover, the absolute-value-repaired legacy generator and the strict generator have 32 Fourier-mode order inversions. Every Bernstein function is increasing, so no monotone scalar subordination

$$A_S = \Phi(A_{L,|\cdot|})$$

exists in the tested class. **[Proven scoped no-go]**

The historical physical formulas also fail a modern provenance test.

- The number $\sin^2 \theta_W = \alpha_{\text{geo}}/12 = \ln 2/3$ is a conditional evaluation of an assumed map; the factor 12 is not obtained from a strict representation, action, or operator observable.
- The historical expression

$$\alpha_{\text{EM}}^{-1} = \frac{\alpha_{\text{geo}}}{2\beta_{\text{tors}}}(1 - \beta_{\text{tors}})$$

is highly sensitive to β_{tors} , omits the phase factor invoked at the start of its old argument, and is not invariant under a change of distance coordinate.

- The hierarchy $G_{\text{eff}}/G_0 = \beta_{\text{tors}}^N$ does not independently predict N ; the historical calculation recovers $N = 19.5$ from a target hierarchy and then rounds to 20, changing the result by a factor ten.
- Planck-scale chains import dimensional SI anchors. A dimensionless kernel can transport a supplied scale but does not create a length, time, mass, or action unit.

These are not objections to using the old formulas as benchmark hypotheses. They are proofs that the formulas are not yet consequences of the kernel.

P295–P308 also produced constructive results:

1. Four shared probes increase the local Stieltjes inverse rank from 6 to 15 and reduce median pole error at relative noise 10^{-4} from 0.496 to 0.150, although uniform stability is still absent.
2. Lean 4.28 machine-checks the general logical theorem behind staged minimization, and 400 exact rational SPD Schur witnesses agree.
3. The strict six-effect current POVM compiles into an exact 72×72 complex-unitary Givens mesh. The exact reconstruction error is 2.07×10^{-15} ; 10^{-3} -radian parameter quantization gives maximum effect error 1.16×10^{-3} .
4. A supplied hidden-rate process has conditional mutual information 0.0010569 nat, whereas the homogeneous Markov process has zero within 4.1×10^{-17} .
5. A joint five-parameter synthetic detector design has full Fisher rank; the gradient RMSE is 0.00906.
6. The strict fingerprint map is locally rank five. Its rare small-ball probability therefore has leading exponent 5; importance sampling gives exponent 5.217 over the tested finite range.
7. Three exact observation times cannot distinguish homogeneous strict heat flow from a positive time-warped generator; off-grid predictions differ.
8. Synthetic interventions raise a hypothetical bridge-coordinate design rank from 4 to 6, but the coordinate and its law were supplied and are not discoveries about nature.
9. The strict heat/Gibbs family possesses a scale orbit, while the old electromagnetic and gravity role formulas fail the corresponding coordinate-gauge test.
10. A free torsor has no invariant point. Pointing makes an equivariant map unique only after the point is supplied.

No independent P241 event bundle or certified hardware reservoir record was accepted. P242 was not run on invented data. No result closes QW-2191, the full legacy-to-strict bridge, physical-role transfer, a dimensional unit, L_{total} , Standard Model dynamics, general relativity, laboratory validation, or a Theory of Everything.

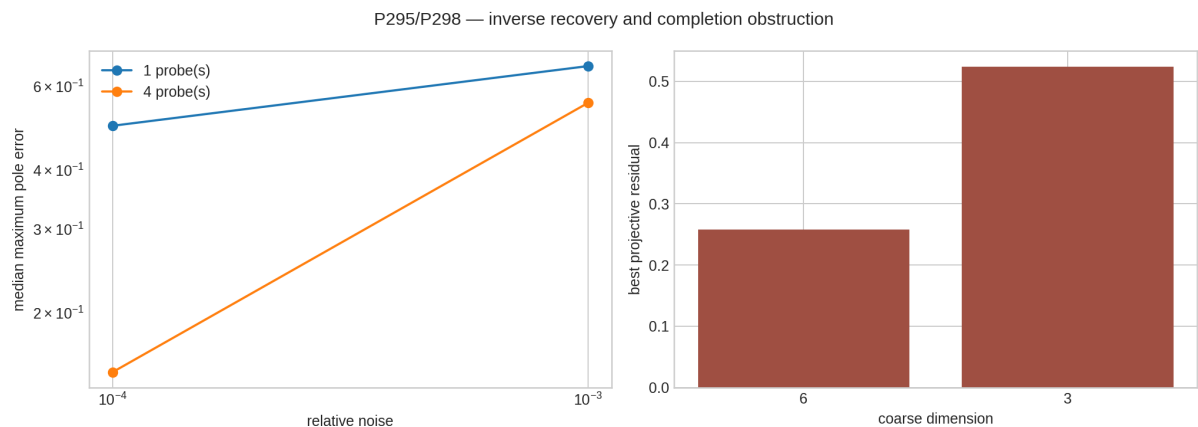


Figure 1.1: Multi-probe inverse recovery and the scalar/Bernstein completion obstruction.

Chapter 2

Scope, ontology, and kernel split

The internal ontology retained in this release is

$$\text{nadsoliton} \longrightarrow \text{light} \longrightarrow \text{matter} \longrightarrow \text{emergent observer}.$$

The nadsoliton is treated within FIN as primordial information in a solitonic state. No additional informational layer is inserted below it. This ontology does not by itself define a laboratory observable.

The historical/intermediate legacy kernel is

$$K_L(d) = \alpha_{\text{geo}} \frac{\cos(\omega_L d + \phi_L)}{1 + \beta_{\text{tors}} d},$$

with the canonical historical tuple

$$\alpha_{\text{geo}} = 4 \ln 2, \quad \omega_L = \frac{\pi}{4}, \quad \phi_L = \frac{\pi}{6}, \quad \beta_{\text{tors}} = 0.01.$$

The strict working kernel is

$$K_S(d) = \frac{\cos(\omega_S d + \phi_S)}{1 + d^\eta}, \quad \omega_S = 0.18575, \quad \phi_S = 0.16250, \quad \eta = 1.8.$$

On Z_{12} , its adjacency W_S and generator are

$$(W_S)_{xy} = K_S(d(x, y)), \quad A_S = \text{diag}(W_S \mathbf{1}) - W_S.$$

The certified finite facts include

$$A_S = A_S^*, \quad A_S \mathbf{1} = 0, \quad A_S \succeq 0,$$

with one zero mode and spectral gap

$$\gamma = 0.7541211542070796.$$

The legacy kernel is an intermediate bridge object, not a co-equal final kernel. The strict kernel may be described as a completed/enriched continuation only through explicit completion evidence. Neither kernel may silently inherit the other kernel's physical roles.

Chapter 3

Reconstruction of the historical legacy-to-physics chain

3.1 What the old construction actually establishes

The historical documentation begins with

$$D_f = 4 \ln 2, \quad \alpha_{\text{geo}} = D_f.$$

The identity $\ln(2^4) = 4 \ln 2$ is exact. The choice of D_f and its identification with a kernel amplitude are structural postulates, not consequences of a spectral theorem.

The multi-component narrative used an architecture of the form

$$K_{\text{total}} = K_{\text{geo}} K_{\text{res}} (1 + 0.2 K_{\text{tors}}) K_{\text{topo}},$$

with representative terms

$$K_{\text{geo}} = e^{-\alpha d}, \quad K_{\text{tors}} = \cos(\omega d + \phi), \quad K_{\text{topo}} \simeq e^{-\beta \Delta n}.$$

The passage from these components and a path sum to

$$\alpha_{\text{geo}} \frac{\cos(\omega d + \phi)}{1 + \beta d}$$

was heuristic. QW1729 found only partial global characteristic closure, while QW1732 found that a hyperbolic envelope can occur in a tuned region. QW1733 rejected a universal path-sum derivation. The correct classification is:

Element	Status
product architecture	model definition
$\omega = \pi/4, \phi = \pi/6$	geometric ansatz
$\beta_{\text{tors}} = 0.01$	calibrated/frozen parameter
path sum $\rightarrow (1 + \beta d)^{-1}$	heuristic, incomplete
final legacy formula	effective intermediate model

3.2 Corrections to the historical diagrams

3.2.1 Cosine zeros

The equation

$$\cos\left(\frac{\pi d}{4} + \frac{\pi}{6}\right) = 0$$

has solutions

$$d = \frac{4}{3} + 4k.$$

There are no integer zeros. Old node lists such as $d = 2, 5, 8, 11$ or $d = 2, 8, 14$ are incorrect.

For the canonical historical kernel:

d	$K_L(d)$
0	2.40113
1	0.710494
2	-1.35911
3	-2.60011
5	-0.683427
7	2.50291
8	2.22327
11	-2.41272

3.2.2 Path-counting exponent

If

$$N(d) \sim d^{1.6}, \quad w_{\text{path}}(d) \sim d^{-\alpha},$$

then

$$K(d) \sim d^{1.6-\alpha}.$$

For $\alpha = 0.6$, this is d^{+1} , not d^{-1} . The stated counting argument would require a path weight of order $d^{-2.6}$ to yield $1/d$.

3.2.3 Inconsistent parameter freezes

Historical materials used several amplitude and damping values in different contexts. A physical prediction can be called blind only after the complete parameter tuple and normalization rule are frozen before the target is revealed.

3.2.4 Simulation is not measurement

Historical Wilson-loop calculations and plots are repository-internal numerical experiments. They are not independent laboratory measurements with apparatus calibration, event-level records, custody separation, and a frozen hold-out.

3.3 Historical physical-role formulas

3.3.1 Electroweak-angle benchmark

$$\sin^2 \theta_W = \frac{\alpha_{\text{geo}}}{12} = \frac{\ln 2}{3} = 0.2310490602.$$

The number follows exactly after accepting the formula. The factor 12 was interpreted as twelve sectors but was not derived from a gauge representation, coupling action, strict spectral projector, or response operator.

One historical algebraic step effectively contains

$$\frac{4}{12} \frac{\alpha_{\text{geo}}}{4/12} = \alpha_{\text{geo}},$$

not $\alpha_{\text{geo}}/12$. Therefore the presented manipulation does not prove the claimed formula.

Status: conditional arithmetic [**Proven**]; kernel-to-observable derivation [**Not established**].

3.3.2 Electromagnetic benchmark

$$\alpha_{\text{EM}}^{-1} = \frac{\alpha_{\text{geo}}}{2\beta_{\text{tors}}} (1 - \beta_{\text{tors}}).$$

At $\beta_{\text{tors}} = 0.01$, this gives

$$\alpha_{\text{EM}}^{-1} = 137.2431417509.$$

Its derivative is

$$\frac{d}{d\beta} \left[\frac{\alpha_{\text{geo}}(1 - \beta)}{2\beta} \right]_{\beta=0.01} = -13862.94 \dots$$

and the reference value 137.035999084 is obtained at approximately $\beta = 0.01001496455$, a change of only 0.15%. This numerical proximity is weak evidence unless β is independently determined.

The old argument invokes

$$K_L(0) = \alpha_{\text{geo}} \cos(\pi/6),$$

but the final formula drops $\cos(\pi/6)$. Thus the result is a separate parameter map rather than a derived evaluation of $K_L(0)$.

Status: conditional arithmetic [**Proven**]; independent prediction [**Refuted by provenance audit**].

3.3.3 Gravity-hierarchy benchmark

$$\frac{G_{\text{eff}}(N)}{G_0} = \beta_{\text{tors}}^N.$$

For $\beta_{\text{tors}} = 0.01$,

$$\beta^{19} = 10^{-38}, \quad \beta^{19.5} = 10^{-39}, \quad \beta^{20} = 10^{-40}.$$

The historical computation used the target 10^{-39} to infer $N = 19.5$, then rounded to 20, whose result differs by a factor ten. The number of layers was therefore not independently predicted.

Status: power-law identity [**Proven**]; gravity law and independent N [**Not established**].

3.3.4 Dimensional scales

The ratio $q = \beta^{-1} = 100$ can transport a supplied scale between layers. It does not determine the initial unit. Historical Planck-scale chains import G_{SI} , c_{SI} , $\hbar_{\text{SI}}$, or a Planck length/time/mass. P3164 independently established the absence of an import-free L, T, M source.

Status: dimensionless scale transport [**Proven, conditional**]; dimension generation [**Refuted on current inputs**].

Chapter 4

Exact legacy–strict decomposition and its limit

Where the legacy cosine is nonzero,

$$K_S(d) = \frac{K_L(d)}{\alpha_{\text{geo}}} \frac{\cos(\omega_S d + \phi_S)}{\cos(\omega_L d + \phi_L)} e^{-C_\eta(d)},$$

where

$$C_\eta(d) = \log \frac{1 + d^\eta}{1 + \beta_{\text{tors}} d}.$$

This identity separates three operations:

1. removal or reinterpretation of the legacy amplitude;
2. phase/frequency transport;
3. nonlinear compression.

It is an algebraic decomposition on a declared support, not a dynamical derivation. The phase ratio becomes singular at continuous legacy zeros and does not define a global natural transformation without additional carrier data. P2377 certifies the compression primitive in its declared scope, while P2378 shows that unit-normalized transport does not source its coupling strength.

A useful finite discriminator is

$$R_{1,7}[K] = \frac{|K(7)|}{|K(1)|},$$

for which

$$R_{1,7}[K_L] = 3.52278, \quad R_{1,7}[K_S] = 0.006708.$$

Legacy has a large long-range oscillatory recurrence; strict is strongly localized. Any completion theorem must explain this qualitative change rather than merely fit one normalization constant.

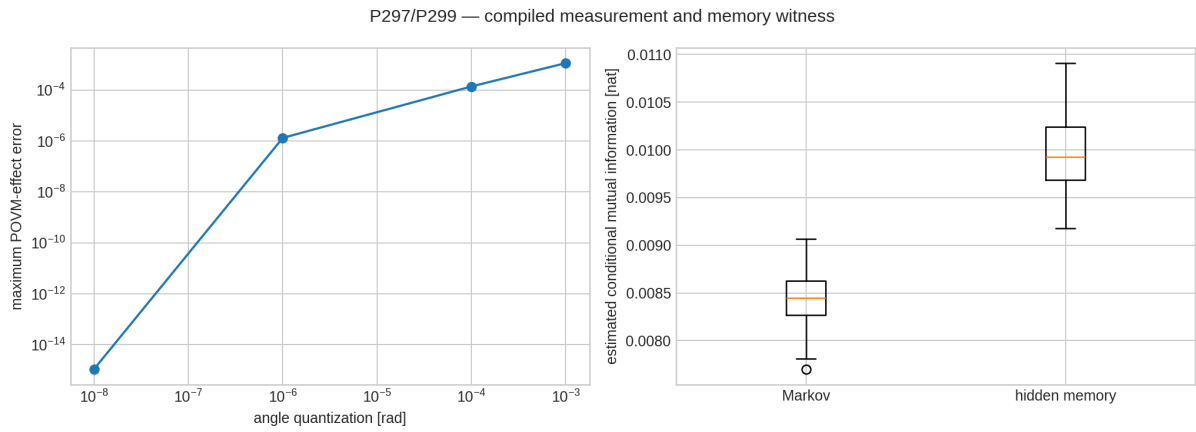


Figure 4.1: Exact complex-unitary current-mesh compilation and a multitime memory witness.

Chapter 5

New theoretical objects O54–O61

5.1 O54 — Spectral Observation Signature

For a declared operational menu $\mathcal{C}$, define

$$\Sigma_{\mathcal{C}}(A) = \left\{ C_r^*(A + zI)^{-1}C_r, p_A^c : r, z, c \in \mathcal{C} \right\}.$$

This object records only the pole/residue and outcome-law information visible to the declared probes, preparations, times, and instruments.

Status: **[Constructed]**. It is injective only under explicit cyclicity, visibility, and full-rank conditions.

5.2 O55 — Bernstein Spectral Completion

The smallest monotone positive scalar subordination candidate is

$$A_S = \Phi(A_L),$$

where, for example,

$$\Phi(\lambda) = b\lambda + \sum_q c_q \frac{\lambda}{\lambda + r_q}, \quad b, c_q, r_q \geq 0.$$

Every such Φ is increasing. P298 finds 32 mode-order inversions between the absolute-value-repaired legacy spectrum and the strict spectrum.

Status: **[Refuted]** for the tested legacy-absolute to strict class.

5.3 O56 — Operational Prediction Pseudometric

$$d_{\mathcal{C}}(K, K') = \sup_{c \in \mathcal{C}} \frac{1}{2} \|p_K^c - p_{K'}^c\|_1.$$

Two kernels may be pointwise different but operationally equivalent relative to a restricted menu. Conversely, a single operational witness can separate them without claiming a pointwise completion.

Status: **[Defined]**; only finite slices have been computed.

5.4 O57 — Positive Path-Scale Measure

$$h_\nu(d) = \int_0^\infty e^{-sd} d\nu(s), \quad \nu \geq 0.$$

The legacy envelope has

$$d\nu_\beta(s) = \frac{1}{\beta} e^{-s/\beta} ds,$$

equivalently the u -integral used above. The strict envelope with $\eta = 1.8$ lies outside this cone.

Status: legacy representation **[Proven]**; strict exclusion **[Proven]**.

This spatial result does not prove temporal non-Markovianity. It only excludes one positive mixture of exponential *distance* attenuations.

5.5 O58 — Scale-Orbit Prediction Quotient

For heat and Gibbs families,

$$(A, t, T) \sim (cA, t/c, cT), \quad c > 0.$$

Projective spectra and rescaled prediction laws live on the quotient. A clock, energy, or temperature standard selects a representative but is not generated by the quotient.

Status: finite invariance **[Proven]**.

5.6 O59 — Role-Transfer Certificate

Define a certificate

$$\text{RT} = (B, R_L, R_S, \tau, \Pi),$$

where B is a typed legacy-to-strict completion, R_L and R_S are role maps, τ is the semantic transport, and Π certifies representation, operational, and scale invariance. The central square must commute:

$$R_S \circ B = \tau \circ R_L.$$

Status: certificate schema **[Constructed]**. No historical electroweak, electromagnetic, or gravity role passes it.

5.7 O60 — Pointed Operational Kernel Package

$$\mathfrak{K}_{\text{phys}} = (K, \rho, A, \mathcal{T}, \mathcal{P}, \mathcal{I}, \mathcal{E}, \mathcal{R}, t_0, \kappa).$$

The entries are:

- kernel K ;
- state ρ ;

- generator A ;
- clock/time family $\mathcal{T}$;
- preparation family $\mathcal{P}$;
- instruments $\mathcal{I}$;
- environment $\mathcal{E}$;
- event record $\mathcal{R}$;
- operational pointing/reference t_0 ;
- dimensional scale κ .

This is the smallest package in this release that can state a physical prediction without silently omitting the observer, apparatus, or unit.

Status: [Mathematically defined]. FIN does not derive all entries.

5.8 O61 — Typed Legacy–Strict Completion Span

$$K_L \xleftarrow{\pi_L} \mathcal{P} \xrightarrow{\pi_S} K_S.$$

The parent $\mathcal{P}$ must keep the following components separately typed:

(radial attenuation, phase resource, compression,
 $\mathcal{P} =$ sector state, orientation/sign, normalization,
 operational map).

A direct sum $A_L \oplus A_S$ is excluded as a trivial parent. A valid span must contain cross-support structure, naturality, or an experimentally distinguishable common mechanism.

Status: [Proposed bridge architecture], not a completed bridge.

Chapter 6

Program results P295–P308

6.1 P295 — Multi-probe operator-valued Stieltjes recovery

For three visible poles,

$$S_r(z) = \sum_{j=1}^3 \frac{w_{rj}}{z + \mu_j}, \quad w_{rj} \geq 0,$$

the poles are shared across probes and the residues are probe dependent.

The true positive poles were

$$(1.1710910, 1.5263637, 1.8883092).$$

Probes	Relative noise	Median max pole error	95% max pole error	Median max residue error
1	10^{-4}	0.4955	0.6617	0.0860
4	10^{-4}	0.1502	0.4778	0.0425
1	10^{-3}	0.6617	0.8081	0.3427
4	10^{-3}	0.5538	0.6617	0.0861

The single-probe Jacobian has rank 6; the four-probe stacked Jacobian has rank 15, equal to the number of fitted parameters in each case.

Verdict: local injectivity [**Proven, conditional**]; frozen recovery [**Strong evidence**]. Full rank does not imply uniform minimax stability.

6.2 P296 — General staged elimination

Let $F(x, y, z)$ take values in a transitive ordered relation. If $z_*(x, y)$ is universally minimal for fixed x, y , and $y_*(x)$ is universally minimal after the inner elimination, then

$$F(x, y_*, z_*(x, y_*)) \leq F(x, y, z)$$

for every y, z .

Lean 4.28 machine-checks this abstract theorem and the uniqueness of an attained minimum value under antisymmetry. Separately, 400 exact rational SPD 3×3 matrices satisfy equality between nested and direct Schur elimination.

Verdict: [\[Proven\]](#). The dependency-free Lean theorem is the general variational logic, not a complete Mathlib theorem for positive matrix blocks.

6.3 P297 — Complex-unitary compilation of the strict current POVM

For six strict current effects E_y ,

$$V = \begin{bmatrix} \sqrt{E_1} \\ \vdots \\ \sqrt{E_6} \end{bmatrix}, \quad V^*V = I_{12}.$$

An orthonormal complement extends V to a 72×72 complex unitary. A complex Givens decomposition uses 1,428 two-mode rotations.

Parameter quantization	Unit. reconstruction error	Max effect error
exact	2.07×10^{-15}	1.11×10^{-15}
10^{-6} rad	3.27×10^{-6}	1.33×10^{-6}
10^{-4} rad	3.06×10^{-4}	1.40×10^{-4}
10^{-3} rad	3.04×10^{-3}	1.16×10^{-3}

All quantized reconstructions remain unitary to numerical precision because the Givens parameters are quantized within a unitary parameterization.

Verdict: ideal compilation [\[Proven\]](#); tolerance response [\[Strong evidence\]](#). This is not a fabricated photonic chip, a loss map, or a detector calibration.

6.4 P298 — Scalar complement and Bernstein completion no-go

A scalar lifted complement

$$\tilde{A} = EBE^* + \alpha(I - EE^*)$$

forces an eigenvalue α of multiplicity at least $12 - m$. The strict generator has maximum eigenvalue multiplicity two. Therefore $m = 6$ and $m = 3$ scalar complements cannot be unitarily equivalent to strict for any α .

The best projective residuals are nonzero, and the legacy-absolute/strict mode comparison has 32 order inversions. Since every Bernstein function is increasing, a monotone scalar functional calculus cannot repair them.

Verdict: exact scalar/Bernstein completion [\[Refuted\]](#). A non-scalar parent, anisotropic source, signed resource, or carrier transport is new structure.

6.5 P299 — Multitime memory ledger

At calibrated step $\tau = 0.35$, compare:

$$P = e^{-\tau A_S}$$

with a persistent hidden-rate mixture using rates 0.65 and 1.35.

Quantity	Result
homogeneous conditional mutual information	-4.09×10^{-17} nat
hidden-memory conditional mutual information	0.00105689 nat
best one-step Markov joint-prediction TV	0.0215679
shots per replication	80,000
frozen classifier accuracy	1.000

Verdict: finite Markov-order witness **[Proven]**; count audit **[Strong evidence]**. The hidden bath and rates were supplied.

6.6 P300 — Positive path-mixture and fixed-phase no-go

On $d = 1, \dots, 6$, the strict weights are positive. The historical legacy cosine is negative at $d = 2, 3, 4, 5$. A positive amplitude and positive denominator preserve those legacy signs; they cannot produce strict.

For

$$h_\eta(d) = \frac{1}{1 + d^\eta},$$

$$h''_\eta(d) = \frac{\eta d^{\eta-2} [(\eta + 1)d^\eta - (\eta - 1)]}{(1 + d^\eta)^3}.$$

When $\eta = 1.8$, this is negative near the origin. The tested value at half the inflection distance is -1.06970 .

The 80-point Gauss–Laguerre verification of the exact legacy mixture has maximum residual 1.21×10^{-14} .

Verdict: positive fixed-phase distance-mixture bridge **[Refuted]**.

6.7 P301 — External P241 intake

The repository was scanned for a non-template bundle manifest, raw event records, custody metadata, and a frozen hold-out. Files with names suggesting external analyses exist in older lanes, but no bundle satisfied the P241 laboratory-event admission contract used here.

Verdict: **[Blocked by external evidence]**. P242 was not executed on synthetic substitutes.

6.8 P302 — Sequential detector design with nuisance tomography

The synthetic categorical click model jointly estimates:

$$\theta = (g, \eta_{\text{det}}, V, d_{\text{dark}}, c_{\text{xtalk}}).$$

Dark, balanced, crosstalk, contrast, and symmetric physics configurations were used. A grid search over displacement h maximized the Fisher log-determinant.

Quantity	Result
optimal tested h	1.375
Fisher rank	5
predicted gradient standard deviation	0.0101464
empirical gradient bias	0.000599
empirical gradient RMSE	0.009063
nominal 95% coverage	0.9833
replications	60

Verdict: [Strong evidence, synthetic]. The result is an experimental design, not a physical calibration.

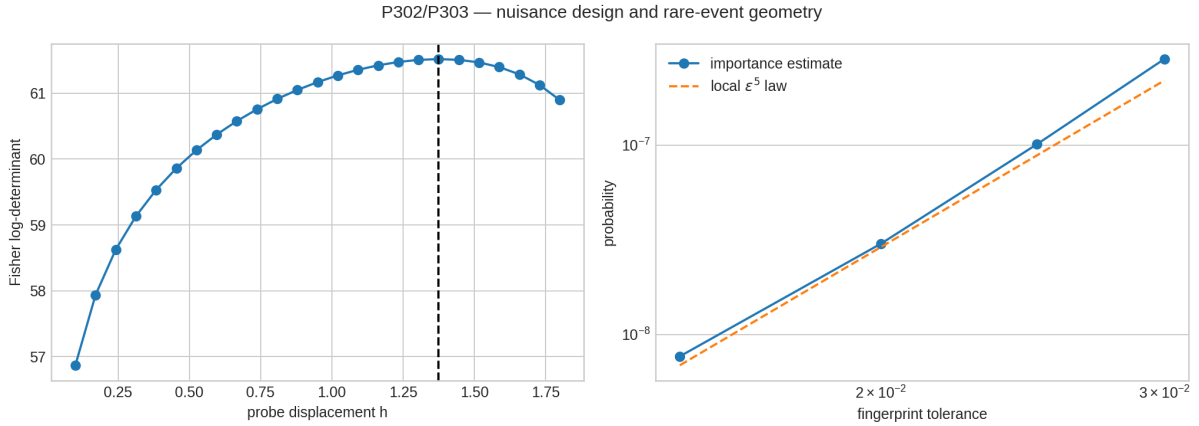


Figure 6.1: Joint detector nuisance design and strict-fingerprint rare-event geometry.

6.9 P303 — Local strict-fingerprint rarity

Positive radial weights normalized to the simplex have five independent coordinates. The derivative of the sorted projective spectral fingerprint at strict has rank five and

$$\det(J^\top J) = 4880.98495.$$

Under the uniform simplex null, the local small-ball law is

$$\Pr(\|F(w) - F(w_S)\| \leq \varepsilon) \sim \frac{5! \operatorname{Vol}(B_5)}{\sqrt{\det(J^\top J)}} \varepsilon^5.$$

Importance sampling over $\varepsilon = 0.015, 0.020, 0.025, 0.030$ gives fitted exponent 5.217.

Verdict: local exponent [**Proven**]; finite-tolerance comparison [**Moderate evidence**]. Singular charts and global look-elsewhere effects remain outside the theorem.

6.10 P304 — Adversarial finite-time equivalence

For observed times $t_i \in \{0.15, 0.35, 0.65\}$, define

$$\tau(t) = t + \epsilon t \prod_i (t - t_i)^2.$$

The selected ϵ gives

$$0.99626 \leq \tau'(t) \leq 1.25$$

on the tested interval, so

$$A_{\text{alt}}(t) = \tau'(t) A_S$$

is a positive time-dependent rate. At every observed time,

$$e^{-\tau(t_i)A_S} = e^{-t_i A_S}$$

exactly, while the maximum off-grid vertex total-variation difference is 0.002684.

Verdict: unrestricted clock/mechanism identification from a finite time grid [**Refuted**]. Homogeneity or a restricted clock family must be an explicit premise.

6.11 P305 — Intervention-assisted bridge-coordinate discovery

The hypothetical coordinate

$$K(\lambda) = (1 - \lambda) \widehat{K}_{L,|\cdot|} + \lambda \widehat{K}_S$$

was supplied, together with the synthetic law

$$\dot{\lambda} = 0.55u - 0.32\lambda - 0.18\lambda^2.$$

Passive data have library rank four. The intervention panel has joint rank six and recovers

$$(0.55070, -0.31999, -0.18054)$$

for the three nonzero coefficients. A new hold-out control has coordinate RMSE 9.48×10^{-6} .

Verdict: identification method [**Strong evidence, synthetic**]; FIN interpretation [**Speculative**]. A coordinate-dependent law is not yet a natural completion theorem.

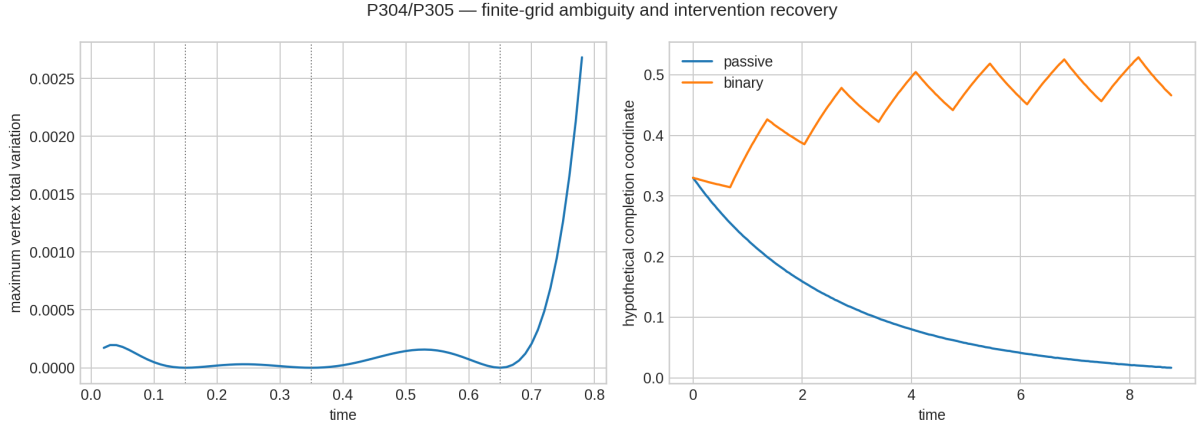


Figure 6.2: Finite-time clock ambiguity and intervention-assisted synthetic completion-coordinate recovery.

6.12 P306 — Physical reservoir evidence gate

No independently certified hardware record was found with a calibrated clock, detector response, raw events, custody separation, and a preregistered hold-out.

Verdict: [Blocked by external evidence]. Reservoir simulations do not become hardware evidence by renaming their outputs.

6.13 P307 — Scale orbit and legacy-role invariance

For $c > 0$,

$$\text{sig}(cA_S) = \text{sig}(A_S),$$

$$e^{-t(cA_S)} = e^{-(ct)A_S},$$

and the Gibbs law is unchanged under

$$(A_S, T) \mapsto (cA_S, cT).$$

The Fisher sensitivity to log generator scale and log clock conversion has rank one, with null direction $(1, -1)$.

Under a distance-coordinate change $d' = cd$, the same legacy denominator is represented by

$$\beta' = \frac{\beta}{c}.$$

The old electroweak number is unchanged because it ignores β , but it does not factor through a strict observable. The old electromagnetic formula varies by a factor 20.39 over the tested coordinate gauges; the β^{20} hierarchy varies by 1.048576×10^{26} .

Verdict: unchanged electromagnetic and gravity role transfer [Refuted]; electroweak transfer [Blocked].

6.14 P308 — Pointed role-transfer theorem

Lean proves:

1. a free nontrivial action has no globally invariant point;
2. two equivariant maps on a transitive torsor are equal if they agree at one supplied point.

Finite cyclic torsors of order 2 through 8 have:

- zero invariant global sections;
- n equivariant self-maps;
- one pointed equivariant map.

None of the historical electroweak, electromagnetic, or gravity roles has all four certificate components: completion naturality, a strict observable, operational/scale invariance, and a dimensional anchor where required.

Verdict: pointed uniqueness **[Proven]**; physical-role transfer **[Blocked]**. QW-2191 remains open.

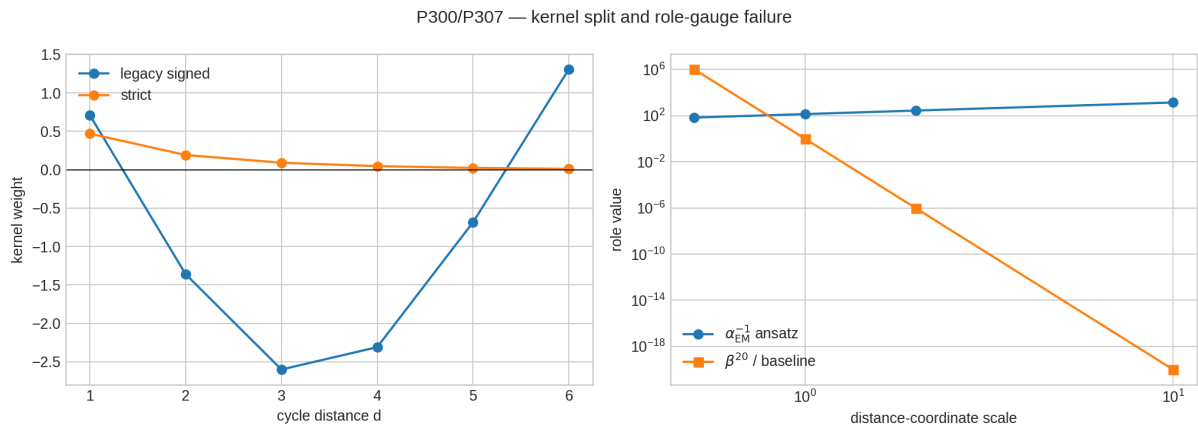


Figure 6.3: The legacy/strict kernel split and the coordinate-gauge failure of historical physical-role formulas.

Chapter 7

Cross-program synthesis

7.1 What survives from the old legacy intuition

Three ideas survive, after correction.

First, the legacy kernel is a useful **long-range oscillatory carrier**. Its hyperbolic attenuation is not arbitrary: it is an exact positive mixture of exponential distance scales.

Second, the old decomposition anticipated that phase, attenuation, topology, and normalization should be separate ingredients. The modern strict analysis confirms that they cannot be collapsed into one scalar parameter.

Third, the historical physical formulas provide sharply defined benchmark maps that can be falsified. Their value is now methodological: they tell us which role-transfer diagrams must commute and which independent measurements would be needed.

7.2 What does not survive

The following inferences fail:

numerical agreement $\nrightarrow$ operator-derived observable,

dimensionless damping parameter $\nrightarrow$ physical unit,

legacy parameter role $\nrightarrow$ strict physical role,

positive path mixture $\nrightarrow$ strict compression,

spectral symmetry $\nrightarrow$ selector or physical pointing.

7.3 The new bridge hypothesis

The smallest bridge still compatible with all no-go results is not a scalar formula $K_S = f(K_L)$. It is a typed completion span

$$K_L \leftarrow \mathcal{P} \rightarrow K_S$$

combined with an operational quotient:

$$\mathcal{P} \longrightarrow \Sigma_{\mathcal{C}}(A) \longrightarrow p(\text{record} \mid \text{preparation, clock, instrument}).$$

At least one of the following must be genuinely new:

- a signed or complex interference measure;
- a non-scalar common parent;
- an anisotropic source;
- a metric/carrier transport;
- an independently prepared operational point;
- a dimensional reference.

This is a necessity classification, not proof that any one candidate is realized.

7.4 Why the old constants must bifurcate

The old α_{geo} simultaneously served as:

- an information/fractal number $4 \ln 2$;
- a kernel amplitude;
- an input to a physical coupling formula.

The old β_{tors} simultaneously served as:

- a radial attenuation coefficient;
- a hierarchy ratio;
- an implicit scale converter.

P307 shows these identifications are not representation safe. The modern theory therefore requires **role bifurcation**:

$$\alpha_{\text{geo}} \rightsquigarrow (\alpha_{\text{info}}, \alpha_{\text{amplitude}}, \alpha_{\text{role}}),$$

$$\beta_{\text{tors}} \rightsquigarrow (\beta_{\text{shape}}, \beta_{\text{scale}}, \beta_{\text{role}}),$$

followed by explicit theorems if any components are to be reidentified.

7.5 Deepest present interpretation

The deepest interpretation surviving this round is:

FIN currently defines a finite spectral-information dynamics with a historically informative but incomplete bridge kernel. Its strict wave/diffusion duality and information-processing constructions are functional-calculus shadows of one positive generator. Turning that generator into physics requires a pointed operational kernel package and a role-transfer certificate. Neither the package nor the certificate follows from the radial kernel alone.

Chapter 8

Failed approaches and impossibility boundaries

Candidate	Result	Exact boundary
positive fixed-phase damping	[Refuted]	four sign mismatches
positive exponential distance mixture	[Refuted for strict]	strict envelope not completely monotone
scalar isotropic complement	[Refuted]	forced flat-band multiplicity
Bernstein scalar functional calculus	[Refuted]	32 mode-order inversions
finite-time unrestricted mechanism ID	[Refuted]	positive clock-warp counterexample
unchanged legacy EM role	[Refuted]	coordinate-gauge dependence
unchanged legacy gravity role	[Refuted]	coordinate-gauge dependence
electroweak role transfer	[Blocked]	no strict observable/factorization
selector from free symmetry	[Refuted]	no invariant torsor point
laboratory confirmation	[Blocked]	no admitted independent event bundle
dimensional emergence	[Blocked]	scale orbit has no internal representative

None of these scoped no-go results proves that no legacy-to-strict bridge or physical theory can exist. They prove that additional structure cannot remain implicit.

Chapter 9

Recommended Programs P309–P322

Probabilities estimate the chance of a decisive result, including a rigorous no-go. They are planning judgments, not statistical posterior probabilities.

Rank	Program	Study	Decisive output	Probability
1	P309	multi-probe Stieltjes minimax theorem	Le Cam/Cramér–Rao lower bound and optimal cyclic probes	0.84
2	P310	full formal Schur theorem	Mathlib proof for arbitrary finite positive block operators	0.82
3	P311	lossy complex-unitary mesh	dilation plus loss, detector confusion, phase drift, and robust distinguishability	0.88
4	P312	nontrivial common-parent SDP	construct or obstruct a cross-supported parent for C_{12}, C_{16}, C_{24}	0.80
5	P313	Mellin/regular-variation bridge audit	classify all positive path-counting laws compatible with the strict tail	0.86
6	P314	phase-resource theorem	test whether legacy torsion phase can naturally generate the strict non-torsion phase	0.83
7	P315	operational pseudometric/comb SDP	optimal preparation, time, and instrument separating legacy and strict	0.78
8	P316	minimal signed path-scale completion	quantify minimum negative/complex measure required to reproduce strict	0.85
9	P317	independent P241 admission	validate a real hash-committed event bundle without synthetic substitution	0.30
10	P318	Bayesian sequential detector drift	nuisance tracking, stopping rule, calibration transfer, and coverage under drift	0.72
11	P319	global coarea/look-elsewhere theorem	stratify eigenvalue-degeneracy charts and bound family-wise false positives	0.69
12	P320	restricted-clock experimental design	choose off-grid times/interventions that defeat the P304 warp class	0.77
13	P321	physical reservoir gate	preregister and execute an external hardware reservoir comparison	0.28
14	P322	single-role naturality certificate	attempt one complete strict electroweak role map; otherwise prove its obstruction	0.61

9.1 Recommended immediate sequence

$$P311 \longrightarrow P316 \longrightarrow P312 \longrightarrow P315 \longrightarrow P320$$

This sequence:

1. converts the ideal P297 measurement into a realistic error model;
2. measures the extra signed/complex resource demanded by P300;
3. tests the smallest nontrivial common parent;
4. replaces pointwise resemblance with operational distinguishability;

5. designs a clock-sensitive falsification of the finite-time ambiguity.

P317 and P321 must remain external gates. They should be executed only after real providers, registrars, calibrated apparatus, event records, and frozen hold-outs exist.

Chapter 10

Reproducibility

10.1 Executable artifacts

- `fin_programs_295_308.py` — deterministic P295–P308 runner;
- `test_fin_programs_295_308.py` — regression and scientific-boundary suite;
- `FIN_Programs_295_308_Formal_Core.lean` — dependency-free Lean 4.28 certificate;
- `FIN_Programs_295_308_Results.json` — machine-readable results;
- `FIN_Programs_295_308_Summary.csv` — program matrix;
- eight detailed CSV tables;
- five figures in `FIN_Programs_295_308_Figures/`.

The random seed is

20260801.

10.2 Verification command

Run `python3 -m unittest -v test_fin_programs_295_308.py` with the environment variable `MPLCONFIGDIR=/tmp/mpl-fin`.

Expected result: **17 tests passed, 0 failed.**

The suite recompiles the Lean certificate.

10.3 Scientific boundary

All detector, memory, adaptation, and finite-count data in this release are synthetic and are labeled as such. The external admission programs explicitly return blocked rather than manufacturing evidence.

Chapter 11

Final answer

The old legacy work and the modern strict analysis do reveal one common mathematical story, but not the old physical one.

The common story is a transition from a positive multiscale oscillatory carrier to a localized positive spectral generator. The old hyperbolic attenuation has an exact path-scale-mixture interpretation. The strict attenuation, phase, and spectrum prove that this interpretation is insufficient. The missing object is therefore not another fitted constant. It is a typed operational completion containing a non-scalar or signed phase resource, a physical pointing, and a dimensional reference.

The deepest valid synthesis is:

$$\begin{array}{c} \text{legacy carrier} \leftarrow \text{typed completion parent} \rightarrow \text{strict spectral generator} \\ \xrightarrow{\mathfrak{K}_{\text{phys}}} \text{testable outcome laws} \end{array}$$

The first two arrows remain incomplete; the last arrow requires operational and dimensional resources that are not generated by the kernel. Historical physical-role formulas may be retained only as preregistered hypotheses until they factor through this diagram and pass the Role-Transfer Certificate.

Chapter 12

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Reproduction certificate

```
MPLCONFIGDIR=/tmp/mpl-fin \  
python3 -m unittest -v test_fin_programs_295_308.py
```

Expected result: 17 tests passed, 0 failed. The suite recompiles the dependency-free Lean 4.28 certificate. Detector, memory, adaptation, and finite-count studies are synthetic. No independent P241 laboratory bundle or certified hardware reservoir record was admitted.

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